Set 29: Student Academic Performance

AIM

To analyze student academic performance using R programming by creating a stacked area chart, a boxplot of study hours grouped by attendance quartiles, and a density plot of test scores.

ALGORITHM

1. Create a dataset containing Student ID, Age, Study Hours, Attendance, Test Score, and Participation Score.
2. Create a stacked area chart to compare Test Score and Participation Score across students.
3. Calculate attendance quartiles and group students according to their attendance levels.
4. Create a boxplot of Study Hours for each attendance quartile.
5. Create a density plot to examine the distribution of Test Scores.
6. Interpret the trends, median study hours, variability, and possible outliers.

PROGRAM

This version uses base R only, so no additional packages are required.

```
student <- data.frame(
  ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)


x <- 1:5
plot(x, student$Test_Score, type = "n",
     ylim = c(0, max(student$Test_Score +
                     student$Participation_Score)),
     xaxt = "n", xlab = "Student ID", ylab = "Score",
     main = "Stacked Area Chart of Student Scores")
axis(1, at = x, labels = student$ID)
```

```
polygon(c(x, rev(x)),
        c(rep(0, 5), rev(student$Test_Score)),
        col = "skyblue")
polygon(c(x, rev(x)),
        c(student$Test_Score,
          rev(student$Test_Score + student$Participation_Score)),
        col = "orange")

legend("topleft",
       legend = c("Test Score", "Participation Score"),
       fill = c("skyblue", "orange"))

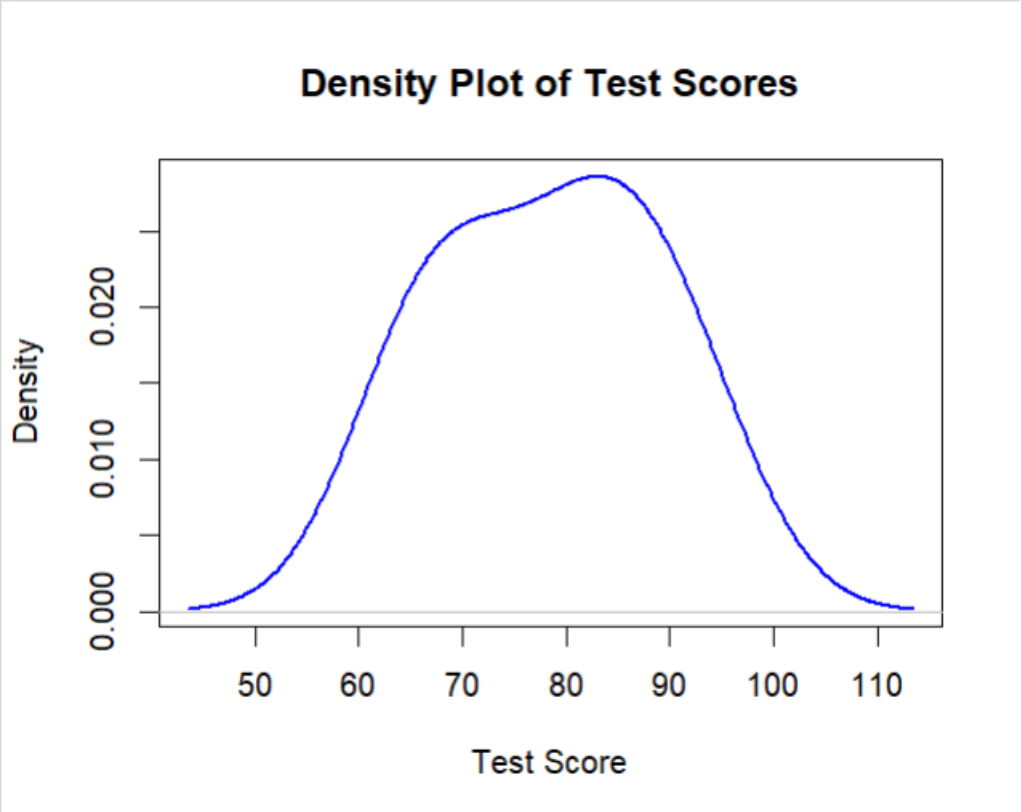
q <- cut(rank(student$Attendance, ties.method = "first"),
         breaks = c(0, 1, 2, 3, 5),
         labels = c("Q1", "Q2", "Q3", "Q4"))

boxplot(student$Study_Hours ~ q,
        col = c("skyblue", "orange", "lightgreen", "pink"),
        xlab = "Attendance Quartile",
        ylab = "Study Hours",
        main = "Study Hours by Attendance Quartile")

plot(density(student$Test_Score),
     main = "Density Plot of Test Scores",
     xlab = "Test Score", col = "blue", lwd = 2)

print(student)
```

OUTPUT:



RESULT

The stacked area chart compares test and participation scores across students. Student S3 has the highest test score (92) and participation score (9). The boxplot compares study hours across attendance groups, while the density plot displays the distribution of test scores.

The data suggests that students with higher attendance often have higher scores in this small sample, although five observations are insufficient to establish a reliable general relationship.